Categories, Proofs, and Processes: Assignment 4

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Question 1

Proposition 1.1. *The covariant Hom-functor $\mathcal{C}(A,-) : \mathcal{C} \to$ **Set** preserves all finite limits.*

Proof. Let $D : \mathcal{J} \to \mathcal{C}$ be a diagram where $\mathcal{J}$ is a finite category. Suppose the cone $(C, (c_1, c_2, ..., c_N))$ to D is a finite limit. By being a cone, for any $f : I \to J$ in $\mathcal{J}$, we have

$$\begin{tikzcd} D\ I \arrow[rr, "D\ f"] & & D\ J \\ & C \arrow[ul, "c_I"] \arrow[ur, "c_J"'] & \end{tikzcd}$$

$$D\ f \circ c_I = c_J$$

We show that $(\mathcal{C}(A,C), (\mathcal{C}(A,c_1), , ..., \mathcal{C}(A,c_N)))$ is a cone and a limit to $\mathcal{C}(A,-) \circ D : \mathcal{J} \to$ **Set**.

$$\mathcal{C} \xrightarrow{\mathcal{C}(A,-)} \mathbf{Set}$$

$$\begin{tikzcd} C \arrow[r, mapsto] \arrow[d, "c_I"'] & \mathcal{C}(A,C) \arrow[d, "\mathcal{C}(A,c_I)"] \\ D\ I \arrow[r, mapsto] \arrow[d, "D\ f"'] & \mathcal{C}(A, D\ I) \arrow[d, "\mathcal{C}(A, D\ f)"] \\ D\ J \arrow[r, mapsto] & \mathcal{C}(A, D\ J) \end{tikzcd}$$

Suppose we have an arbitrary $f : I \to J$ in $\mathcal{J}$, so we get

$$\mathcal{C}(A, (D\ f)) : \mathcal{C}(A, (D\ I)) \to \mathcal{C}(A, (D\ J))$$

Consider the morphisms $D\ I \xleftarrow{c_I} C \xrightarrow{c_J} D\ J$, so

$$\mathcal{C}(A, c_I) : \mathcal{C}(A, C) \to \mathcal{C}(A, (D\ I)) := h \mapsto c_I \circ h$$

$$\mathcal{C}(A, c_J) : \mathcal{C}(A, C) \to \mathcal{C}(A, (D\ J)) := h' \mapsto c_J \circ h'$$

$$\begin{tikzcd}
\mathcal{C}(A, D\ I) \arrow[rr, "\mathcal{C}(A,D\ f)"] & & \mathcal{C}(A, D\ J) \\
& \mathcal{C}(A, C) \arrow[ul, "\mathcal{C}(A,c_I)"] \arrow[ur, "\mathcal{C}(A,c_J)"'] &
\end{tikzcd}$$

Then the diagram commutes since

$$\begin{aligned}
\mathcal{C}(A, (D\ f)) \circ \mathcal{C}(A, c_I) &= (h \mapsto (D\ f) \circ h) \circ (h' \mapsto c_I \circ h') \\
&= h \mapsto (D\ f) \circ c_I \circ h \\
&= h \mapsto c_J \circ h \\
&= \mathcal{C}(A, c_J)
\end{aligned} \tag{1}$$

So $(\mathcal{C}(A, C), (\mathcal{C}(A, c_1), , ..., \mathcal{C}(A, c_N)))$ is a cone to $\mathcal{C}(A, -) \circ D$.

To show the limit, consider $\mathcal{C}(A, C)$ in **Set** and let (B, γ) be some other cone to $\mathcal{C}(A, -) \circ D$, where γ is

$$\{\gamma_I : B \to \mathcal{C}(A, (D\ I))\}_{I \in \text{Ob}(\mathcal{J})}$$

Since $(C, (c_1, c_2, ..., c_N))$ is a limit to D, for any cone (A, δ) to D, there is a unique arrow $a : A \to C$ such that $\forall I \in \text{Ob}(\mathcal{J}), c_I \circ a = \delta_I$.

Note that $\forall b \in B, \gamma_I(b) \in \mathcal{C}(A, (D\ I))$. Then for each $b \in B$, we have the cone

$$(A, \{\gamma_I(b) : A \to D\ I\}_{I \in \text{Ob}(\mathcal{J})})$$

From being the limit, there is a unique map $u_b : A \to C$ such that

$$\begin{tikzcd}
& D\ I \\
A \arrow[ur, "\gamma_I(b)"] \arrow[r, "u_b"'] & C \arrow[u, "c_I"']
\end{tikzcd}$$

$$\forall I \in \text{Ob}(\mathcal{J}), c_I \circ u_b = \gamma_I(b)$$

By defining the arrow $g : B \to \mathcal{C}(A, C)$ such that $\forall b \in B, g(b) = u_b$, g is the unique arrow from the cone (B, γ) to $(\mathcal{C}(A, C), (\mathcal{C}(A, c_1), , ..., \mathcal{C}(A, c_N)))$, determined by the uniqueness of each u_b.

$$\begin{tikzcd}
& \mathcal{C}(A, D\ I) \\
B \arrow[ur, "\gamma_I"] \arrow[r, "g"'] & \mathcal{C}(A, C) \arrow[u, "\mathcal{C}(A,c_I)"']
\end{tikzcd}$$

Thus, $(\mathcal{C}(A, C), (\mathcal{C}(A, c_1), , ..., \mathcal{C}(A, c_N)))$ is the finite limit to $\mathcal{C}(A, -) \circ D$.

□

Corollary 1.2. *Since all representable functors are naturally isomorphic to the covariant Hom-functor at A, $\mathcal{C}(A, -) : C \to$ **Set** for some A, representable functors then preserve finite limits.*

Question 2

Proposition 2.3. *The natural transformation $\Delta : Id \to F$ given by arrows $\Delta_X : X \to X \times X := x \mapsto (x, x)$ is the only natural transformation of the type $Id \to F$.*

Proof. Δ is a natural transformation since the following naturality square commutes, where $f : A \to B$ is an arbitrary arrow in **Set**, and $f(a) = b \in B$, with the mappings as shown, for every $a \in A, A \in \mathrm{Ob}(\mathbf{Set})$.

$$\begin{tikzcd}
\mathrm{Id} \arrow[ddd, "\Delta"'] & \mathrm{Id}\ A = A \arrow[ddd, "\Delta_A"'] \arrow[rrr, "\mathrm{Id}\ f=f"'] & & & \mathrm{Id}\ B = B \arrow[ddd, "\Delta_B"'] \\
& & a \arrow[r, mapsto] \arrow[d, mapsto] & b \arrow[d, mapsto] & \\
& & (a,a) \arrow[r, mapsto] & (b,b) & \\
F & F\ A = A \times A \arrow[rrr, "F\ f=f\times f"] & & & F\ B = B \times B
\end{tikzcd}$$

Suppose $\Gamma : \mathrm{Id} \to F$ is another natural transformation. Then for any function $f : \{*\} \to A$ on the singleton set where $f(*) = a \in A$, the following diagram commutes.

$$\begin{tikzcd}
\mathrm{Id} \arrow[ddd, "\Gamma"'] & \mathrm{Id}\ \{*\} = \{*\} \arrow[ddd, "\Gamma_{\{*\}}"'] \arrow[rrr, "\mathrm{Id}\ f=f"] & & & \mathrm{Id}\ A = A \arrow[ddd, "\Gamma_A"'] \\
& & * \arrow[r, mapsto] \arrow[d, mapsto] & a \arrow[d, mapsto] & \\
& & (*,*) \arrow[r, mapsto] & (a,a) & \\
F & F\ \{*\} := \{*\} \times \{*\} \arrow[rrr, "F\ f=f\times f"'] & & & F\ A = A \times A \\
& = \{(*,*)\} & & &
\end{tikzcd}$$

In other words, we must have

$$\forall A \in \mathrm{Ob}(\mathbf{Set}) : \Gamma_A(a) = (f \times f)(*, *) = (f(*), f(*)) = (a, a)$$

Then $\forall A, \Gamma_A = \Delta_A$, so Δ is the unique natural transformation. $\square$

Question 3

Define

$$\texttt{eval} : \mathcal{D}^{\mathcal{C}} \times \mathcal{C} \to \mathcal{D}$$

$$\texttt{eval} :: (F : \mathcal{C} \to \mathcal{D}, C) \mapsto F\ C, \quad (\Delta : F \to G, h : C \to C') \mapsto (k : F\ C \to G\ C')$$

$$\mathcal{D}^{\mathcal{C}} \times \mathcal{C} \xrightarrow{\texttt{eval}} \mathcal{D}$$

$$\begin{tikzcd}
(F : \mathcal{C} \to \mathcal{D}, C) \arrow[r, mapsto] \arrow[d, "(\Delta:F\to G, h:C\to C')"'] & F\ C \arrow[d, "k:F\ C\to G\ C'"] \\
{} \arrow[r, mapsto] & {} \\
(F : \mathcal{C} \to \mathcal{D}, C') \arrow[r, mapsto] & G\ C'
\end{tikzcd}$$

Lemma 3.4. *$\texttt{eval} : \mathcal{D}^{\mathcal{C}} \times \mathcal{C} \to \mathcal{D}$ is a functor.*

Proof. We use the lemma to show that eval is a functor. Fix an object $F : \mathcal{C} \to \mathcal{D}$ in $\mathcal{D}^{\mathcal{C}}$ so

$$\texttt{eval}(F, -) : \mathcal{C} \to \mathcal{D}$$

$$\texttt{eval}(F, -) :: C \mapsto \texttt{eval}(F, C), \quad f \mapsto F\ f$$

Clearly each object C in $\mathcal{C}$ has a mapping to $F\ C$ in $\mathcal{D}$. For arrow $f : A \to B, g : B \to C$ in $\mathcal{C}$,

$$\begin{aligned}
\texttt{eval}(F, g \circ f) &= F(g \circ f) \\
&= F\ g \circ F\ f \quad \text{(by functoriality)} \\
&= \texttt{eval}(F, g) \circ \texttt{eval}(F, g)
\end{aligned} \tag{2}$$

$$\texttt{eval}(F, 1_A) = F\ 1_A = 1_{F\ A} = 1_{\texttt{eval}(F,A)}$$

So $\texttt{eval}(F, -)$ satisfies functoriality.

Now fix an object C in $\mathcal{C}$ so

$$\texttt{eval}(-, C) : \mathcal{D}^{\mathcal{C}} \to \mathcal{D}$$

$$\texttt{eval}(-, C) :: F \mapsto \texttt{eval}(F, C), \quad t \mapsto t_C$$

Each functor $F : \mathcal{C} \to \mathcal{D}$ in $\mathcal{D}^{\mathcal{C}}$ has a mapping to $F\ C$ in $\mathcal{D}$. For each natural transformation $\Delta : F \to G, \Gamma : G \to H$ in $\mathcal{D}^{\mathcal{C}}$,

$$\begin{tikzcd}
F\ C \arrow[r, "\Delta_C"] \arrow[rr, bend left, "(\Gamma\circ\Delta)_C"] & G\ C \arrow[r, "\Gamma_C"] & H\ C
\end{tikzcd}$$

$$\begin{aligned}
\texttt{eval}(\Gamma \circ \Delta, C) &= (\Gamma \circ \Delta)_C : F\ C \to H\ C \\
&= (\Gamma_C : G\ C \to H\ C) \circ (\Delta_C : F\ C \to H\ C) \\
&= \texttt{eval}(\Gamma, C) \circ \texttt{eval}(\Delta, C)
\end{aligned} \tag{3}$$

$$\texttt{eval}(1_F, C) = 1_{F\ C} = 1_{\texttt{eval}(F,C)}$$

So $\texttt{eval}(-, C)$ is a functor, and the first part of the lemma is satisfied.

Suppose we have a natural transformation $\Delta : F \to G$ in $\mathcal{D}^{\mathcal{C}}$, and an arrow $f : C \to C'$ in $\mathcal{C}$.

$$\begin{tikzcd}
\texttt{eval}(F,C) \arrow[r, "\texttt{eval}(F,f)"] \arrow[d, "\texttt{eval}(\Delta,C)"'] & \texttt{eval}(F,C') \arrow[d, "\texttt{eval}(\Delta,C')"] \\
\texttt{eval}(G,C) \arrow[r, "\texttt{eval}(G,f)"] & \texttt{eval}(G,C')
\end{tikzcd}$$

Then the diagram commutes since the naturality square commutes with Δ being a natural transformation.

$$\begin{tikzcd}
F\ C \arrow[r, "F\ f"] \arrow[d, "\Delta_C"'] & F\ C' \arrow[d, "\Delta_{C'}"] \\
G\ C \arrow[r, "G\ f"] & G\ C'
\end{tikzcd}$$

$$\begin{aligned}
\texttt{eval}(F,f) \circ \texttt{eval}(\Delta, C') &= (F\ f) \circ \Delta'_C \\
&= (G\ f) \circ \Delta_C \quad \text{(by naturality)} \\
&= \texttt{eval}(G,f) \circ \texttt{eval}(\Delta, C)
\end{aligned} \tag{4}$$

The second part of the lemma is satisfied, and $\texttt{eval} : \mathcal{D}^{\mathcal{C}} \times \mathcal{C} \to \mathcal{D}$ is a functor. □

Let $G : \mathcal{E} \times \mathcal{C} \to \mathcal{D}$ be an arbitrary functor. Define

$$\tilde{G} : \mathcal{E} \to \mathcal{D}^{\mathcal{C}}$$

$$\tilde{G} :: A \mapsto G(A, -), \quad (f : A \to B) \mapsto (\Delta : G(A,-) \to G(B,-))$$

where $G(A,-) : \mathcal{C} \to \mathcal{D}$ is a functor for each A in $\mathcal{E}$.

Lemma 3.5. *For a given $G : \mathcal{E} \times \mathcal{C} \to \mathcal{D}$, $\tilde{G} : \mathcal{E} \to \mathcal{D}^{\mathcal{C}}$ is the unique functor where the following diagram commutes.*

$$\begin{tikzcd}
\mathcal{D}^{\mathcal{C}} \times \mathcal{C} \arrow[r, "eval"] & \mathcal{D} \\
\mathcal{E} \times \mathcal{C} \arrow[u, "\tilde{G}\times 1_{\mathcal{C}}"] \arrow[ur, "G"'] &
\end{tikzcd}$$

Proof. Any object A in $\mathcal{E}$ will get mapped to $G(A,-)$ in $\mathcal{D}^{\mathcal{C}}$. Suppose $A \xrightarrow{f} A' \xrightarrow{g} A''$ in $\mathcal{E}$. Then

$$\begin{aligned}
\forall C \in \mathrm{Ob}(\mathcal{C}), ((\tilde{G}\ g) \circ (\tilde{G}\ f))(C) &= (\tilde{G}\ g)(C) \circ (\tilde{G}\ f)(C) \\
&= (\Delta'_C : G(A',C) \to G(A'',C)) \circ (\Delta_C : G(A,C) \to G(A',C)) \\
&= (\Delta' \circ \Delta)_C : G(A,C) \to G(A'',C) \\
&= \tilde{G}(g \circ f)(C)
\end{aligned}$$

$$\tilde{G}\ 1_A = 1_{G(A,-)} = 1_{\tilde{G}\ A} \tag{5}$$

So $\tilde{G} : \mathcal{E} \to \mathcal{D}^{\mathcal{C}}$ is a functor.

Let (E,C) be an object in $\mathcal{E} \times \mathcal{C}$.

$$\texttt{eval} \circ (\tilde{G} \times 1_{\mathcal{C}})(E,C) = \texttt{eval}(G(E,-),C) = G(E,C)$$

Let (f, g) be an arrow in $\mathcal{E} \times \mathcal{C}$.

$$\begin{aligned} \texttt{eval} \circ (\tilde{G} \times 1_{\mathcal{C}})(f : A \to A', g : C \to C') &= \texttt{eval}(\Delta : G(A, -) \to G(A', -), g : C \to C') \\ &= k : G(A, C) \to G(A', C') \\ &= G(f : A \to A', g : C \to C') \end{aligned} \tag{6}$$

So $\texttt{eval} \circ (\tilde{G} \times 1_{\mathcal{C}}) = G$.

To show uniqueness, suppose there is some other functor $H : \mathcal{E} \to \mathcal{D}^{\mathcal{C}}$ such that $\texttt{eval} \circ (H \times 1_{\mathcal{C}}) = G$. Then for each object (E, C) in $\mathcal{E} \times \mathcal{C}$,

$$\texttt{eval} \circ (H \times 1_{\mathcal{C}})(E, C) = \texttt{eval}(H(E), C) = H(E)(C) = G(E, C)$$

So $\forall (E, C) \in \mathrm{Ob}(\mathcal{E} \times \mathcal{C}), H(E)(C) = G(E, C)$. Then for each arrow (f, g) in $\mathcal{E} \times \mathcal{C}$,

$$\begin{aligned} \texttt{eval} \circ (H \times 1_{\mathcal{C}})(f : A \to A', g : C \to C') &= \texttt{eval}(H\ f : H\ A\ \to H\ A', g : C \to C') \\ &= k : H\ A\ C \to H\ A'\ C' \\ &= G(f : A \to A', g : C \to C') \\ &= k : G(A, C) \to G(A', C') \end{aligned} \tag{7}$$

So $\forall (f, g) \in \mathrm{Ob}(\mathcal{E} \times \mathcal{C}), H(f)(g) = G(f, g)$. Thus, $H = \tilde{G}$ is unique. □

Proposition 3.6. *Func$(\mathcal{C}, \mathcal{D})$ is an exponential object in **Cat**.*

Proof. By Lemmas 3.4, 3.5, there is an object $\mathcal{D}^{\mathcal{C}}$ and morphism `eval`: $\mathcal{D}^{\mathcal{C}} \times \mathcal{C} \to \mathcal{D}$, where for every $G : \mathcal{E} \times \mathcal{C} \to \mathcal{D}$, there is a unique morphism $\tilde{G} : \mathcal{E} \to \mathcal{D}^{\mathcal{C}}$ such that $\texttt{eval} \circ (\tilde{G} \times 1_{\mathcal{C}}) = G$. Func$(\mathcal{C}, \mathcal{D})$ is an exponential object in **Cat**. □

Question 4

(a)

Proposition 4.7. *$a \Rightarrow b$ is an exponential.*

Proof.

$$\texttt{eval} : (a \Rightarrow b) \wedge a \leq b$$

Let $f : c \wedge a \leq b$ be an arrow. We want to show the unique arrow $\tilde{f} : c \leq (a \Rightarrow b)$ such that

$$\texttt{eval} \circ (\tilde{f} \times 1_a) = f$$

$$\begin{aligned} \texttt{eval} \circ (\tilde{f} \times 1_a) &= \texttt{eval} \circ (c \leq (a \Rightarrow b) \wedge a \leq a) \\ &= ((a \Rightarrow b) \wedge a \leq b) \circ (c \wedge a \leq (a \Rightarrow b) \wedge a) \\ &= c \wedge a \leq b \\ &= f \end{aligned} \tag{8}$$

Since B is a poset, there is at most 1 arrow between objects - $\tilde{f}$ is a unique arrow.

$a \Rightarrow b$ is an exponential.

□

(b)

$$\perp = \emptyset$$

$$\top = X$$

(c)

The atoms are the elements of the set X as sets. i.e. the set $\{x\}$ where $x \in X$.

(d)

Proposition 4.8. *$t : Id \to At \circ \mathcal{P}^B$ is a natural isomorphism.*

$$
\begin{tikzcd}
\mathrm{Id} \arrow[d, shift right, "t"'] \\
At \circ \mathcal{P}^B \arrow[u, shift right, "t^{-1}"']
\end{tikzcd}
\qquad
\begin{tikzcd}[column sep=huge, row sep=huge]
A \arrow[rr, "f"'] \arrow[d, shift right, "t_A"'] & & B \arrow[d, shift right, "t_B"'] \\
\{\{a\} | a \in A\} \arrow[u, shift right, "t_A^{-1}"'] \arrow[rr, "\{a\} \mapsto \{f(a)\}"] & & \{\{b\} | b \in B\} \arrow[u, shift right, "t_B^{-1}"']
\end{tikzcd}
$$

$$
\begin{tikzcd}
a \arrow[r, mapsto] \arrow[d, mapsto] & f(a) \arrow[d, mapsto] \\
\{a\} \arrow[r, mapsto] & \{f(a)\}
\end{tikzcd}
$$

Proof. To prove the naturality square commutes, we show that for all finite sets A, B,

$$((At \circ \mathcal{P}^B)\ f) \circ t_A = t_B \circ f$$

$$\mathbf{FinSet} \xrightarrow{\mathcal{P}^B} \mathbf{FinBool}^{op} \xrightarrow{At} \mathbf{FinSet}$$

$$
\begin{tikzcd}
A \arrow[r, mapsto] \arrow[d, "f"'] & \mathcal{P}(A) \arrow[r, mapsto] & \{\{a\} | a \in A\} \arrow[d, "At(h)"] \\
B \arrow[r, mapsto] & \mathcal{P}(B) \arrow[u, "h(T)=\{a \in A | f(a) \in T\}"'] \arrow[r, mapsto] & \{\{b\} | b \in B\}
\end{tikzcd}
$$

Note that by definition of the functors $\mathcal{P}^B$ and At, where $\mathcal{P}^B\ f = h : \mathcal{P}(B) \to \mathcal{P}(A)$,

$$
\begin{aligned}
\forall a \in A, (\forall T \in \mathcal{P}(B), f(a) \in T \iff a \in h(T) \iff \{a\} \subseteq h(T)) \\
\iff (\exists! \{b'\} \in \mathcal{P}(B) : \{b'\} \subseteq T \iff b' \in T)
\end{aligned}
\tag{9}
$$

In other words, for all elements $a \in A$, let its corresponding atom $\{a\} \subseteq h(T)$ for all subsets $T \subseteq B$ where $f(a) \in T$, so that $\{b'\} \subseteq B$ is the unique atom that satisfies the property of a Boolean algebra homomorphism.

It must be that $b' = f(a)$ since $\{f(a)\} \subseteq B$ is one such atom that satisfies the property, and by uniqueness. So $\forall a \in A, At(h)(\{a\}) = \{f(a)\} \subseteq B$. t is a natural transformation.

To show the natural isomorphism, we can define the inverse for all finite sets A,

$$t_A^{-1} : \{\{a\} | a \in A\} \to A$$

$$t_A^{-1} :: \{a\} \mapsto a, \quad (k : \{a\} \mapsto \{b\}) \mapsto (f(a) = b)$$

so $t_A^{-1} \circ t_A = 1_{\mathrm{Id}\ A}, t_A \circ t_A^{-1} = 1_{At \circ \mathcal{P}^B\ A}$. We have for all A, t_A is an isomorphism, so t is a natural isomorphism.

□